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Name of Faculty: Dr. Raish Muhammad

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UNIT-II (COORDINATE GEOMETRY OF THREE DIMENSIONS)
LECTURE NO.:- 11

SPHERE

Definition: A sphere is the locus of a point which moves so that its distance from a fixed point is constant. The fixed point is called the centre of the sphere and constant distance the radius.

EQUATION OF A SPHERE

To find the equation of sphere with centre at (a, b, c) and radius r.
Let $P(x, y, z)$ be any point on the sphere.
Then $\sqrt{[(x-a)^2 + (y-b)^2 + (z-c)^2]} = r$.
Or $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$. .... (1)
This is the required equation of the sphere.
The following characteristics of the equation of a sphere can be noted from (1) above:
1. It is of second degree in x, y, z.
2. The coefficients of x^2, y^2, z^2 are each of equal to unity.
3. The product terms of xy, yz, zx are absent.
4. The equation of sphere whose centre is origin (0, 0, 0) is $x^2 + y^2 + z^2 = r^2$.

GENERAL EQUATION OF A SPHERE

The general equation of a sphere is
$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$. .... (2)
It can be written as
$(x+u)^2 + (y+v)^2 + (z+w)^2 = u^2 + v^2 + w^2 - d$.
On comparing it with (1), we see that above equation represents a sphere with centre at $(-u, -v, -w)$ and radius $r = \sqrt{(u^2 + v^2 + w^2 - d)}$.

DIAMETER FORM OF THE EQUATION OF A SPHERE

To find the equation of the sphere described on the line joining the point $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ as diameter.
Let $P(x, y, z)$ be any point on the sphere, described on AB as diameter. The direction ratios of the line AP and BP are respectively $x - x_1, y - y_1, z - z_1$; $x - x_2, y - y_2, z - z_2$.
Angle APB being an angle in a semicircle is equal to right angle.
Therefore $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$.
This is the equation of the sphere with (x_1, y_1, z_1) and (x_2, y_2, z_2) as extremities of a diameter.

THE SPHERE THROUGH FOUR POINTS

The general equation of a sphere is
$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$.
It contains four arbitrary constants u, v, w, d and as such four conditions, each giving rise to one linear relation between the constants are required to uniquely determine a sphere.
To find the equation of a sphere passing through four points (x_i, y_i, z_i), $i = 1, 2, 3, 4$. Let the equation of the sphere be
$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$. .... (1)
Then we have
$x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1 + d = 0$ .... (2)
$x_2^2 + y_2^2 + z_2^2 + 2ux_2 + 2vy_2 + 2wz_2 + d = 0$ .... (3)
$x_3^2 + y_3^2 + z_3^2 + 2ux_3 + 2vy_3 + 2wz_3 + d = 0$ .... (4)
$x_4^2 + y_4^2 + z_4^2 + 2ux_4 + 2vy_4 + 2wz_4 + d = 0$ .... (5)

Equations (2) to (5) are four linear simultaneous equations in u, v, w and d, and as such can be solved to obtain the values of the unknowns u, v, w and d. Substituting these values in equation (1) to get required equation of the sphere.

SOLVED PROBLEMS

Example 1. Find the equation of the sphere through the four points (0, 0, 0), $(-a, b, c)$, (a, b, c), $(a, b, -c)$ and determine its radius.

Solution. Suppose the equation of the sphere passing through the given 4 points is

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0. \quad \text{.... (1)}$$

(1) passes through (0, 0, 0) $\Rightarrow d = 0$. .... (2)

$(-a, b, c)$ lies on (1)

$$\Rightarrow \quad a^2 + b^2 + c^2 - 2ua + 2vb + 2wc = 0 \quad \text{...(3)}$$

$(a, -b, c)$ lies on (1)

$$\Rightarrow \quad a^2 + b^2 + c^2 + 2ua - 2vb + 2wc = 0 \quad \text{.... (4)}$$

$(a, b, -c)$ lies on (1)

$$\Rightarrow \quad a^2 + b^2 + c^2 + 2ua + 2vb - 2wc = 0 \quad \text{.... (5)}$$

We next solve (3), (4), (5) for u, v, w.

(3) – (4) gives $-4ua + 4vb = 0 \Rightarrow ua = vb$ .... (6)

(4) – (5) gives $-4vb + 4wc = 0 \Rightarrow vb = wc$ .... (7)

Put $u = \dfrac{bv}{a}$, $w = \dfrac{b}{c}v$ in (3) to get

$$a^2 + b^2 + c^2 - 2bv + 2bv + 2bv = 0 \Rightarrow v = -\frac{1}{2b}(a^2 + b^2 + c^2) \text{....(8)}$$

$$\therefore \quad u = \frac{-1}{2a}(a^2 + b^2 + c^2),\ v = -\frac{1}{2b}(a^2 + b^2 + c^2),$$

$$w = \frac{-1}{2c}(a^2 + b^2 + c^2)$$

Equation of the required sphere is

$$x^2 + y^2 + z^2 - (a^2 + b^2 + c^2)\left(\frac{x}{a} + \frac{y}{b} + \frac{z}{c}\right) = 0. \quad \underline{\text{Ans.}}$$

Its radius $= \sqrt{(a^2 + b^2 + c^2)^2\left(\frac{1}{4a^2} + \frac{1}{4b^2} + \frac{1}{4c^2}\right)}$

$$= \frac{1}{2}(a^2 + b^2 + c^2)\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right)}. \quad \underline{\text{Ans.}}$$

Example 2. A variable plane passes through a fixed point (a, b, c) and cuts the coordinate axes at A, B, C. Show that the locus of the centre of the sphere OABC is $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$.

Solution. Suppose the equation of the plane is

$$\frac{x}{\lambda} + \frac{y}{\mu} + \frac{z}{\upsilon} = 1, \quad \text{.....(1)}$$

so that $A(\lambda, 0, 0)$, $B(0, \mu, 0)$, $C(0, 0, \upsilon)$

and
$$\frac{a}{\lambda} + \frac{b}{\mu} + \frac{c}{\upsilon} = 1. \quad \text{.....(2)}$$

The equation of the sphere $OABC$ is

$$x^2 + y^2 + z^2 - \lambda x - \mu x - \upsilon z = 0 \quad \text{....(3)}$$

with centre $\left(\frac{1}{2}\lambda, \frac{1}{2}\mu, \frac{1}{2}\upsilon\right)$.

If (α, β, γ) be the centre of the sphere.

$$\alpha = \frac{\lambda}{2}, \beta = \frac{\mu}{2}, \gamma = \frac{\upsilon}{2}$$

i.e. $\lambda = 2\alpha, \mu = 2\beta, \upsilon = 2\gamma.$ ....(4)

(2) then gives

$$\frac{a}{2\alpha} + \frac{b}{2\beta} + \frac{c}{2\gamma} = 1$$

and the locus of (α, β, γ) is

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2.$$

Example 3. A sphere of constant radius k passes through the origin O and cuts the axes in A, B. C. Find the locus of the foot of perpendicular from O to the plane ABC.

Solution. Let the equation of the sphere is

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0. \quad ....(1)$$

Obviously $d = 0$.

Also, $u^2 + v^2 + w^2 = k^2$ ..... (2)

The sphere (1) meets the axes in A, B, C.

$\therefore$ A $[-2u, 0, 0]$, B $[0, -2v, 0]$, c $[0, 0, -2w]$.

Equation of the plane ABC is

$$\frac{x}{-2u} + \frac{y}{-2v} + \frac{z}{-2w} = 1. \quad ....(3)$$

Let P be the foot of the $\perp$ drawn from O to plane (3).

Equations of OP are

$$\frac{x-0}{\frac{-1}{2u}} = \frac{y-0}{\frac{-1}{2v}} = \frac{z-0}{\frac{-1}{2w}} (=r), \text{ say} \quad .....(4)$$

Any point on this line is

$$\left(\frac{-r}{2u}, \frac{-r}{2v}, \frac{-r}{2w}\right).$$

Let P be this point, i.e., P $\left(\frac{-r}{2u}, \frac{-r}{2v}, \frac{-r}{2w}\right)$ for some r.

It lies on the plane (3).

$$\therefore \quad \frac{r}{4u^2} + \frac{r}{4v^2} + \frac{r}{4w^2} = 1$$

$$\therefore \quad r = \frac{4}{(u^{-2} + v^{-2} + w^{-2})}$$

$$P\left(\frac{-r}{2u}, \frac{-r}{2v}, \frac{-r}{2w}\right), r = \frac{4}{(\Sigma u^{-2})}$$

To find the locus of P, let P be (α, β, γ).

$$\alpha = \frac{-r}{2u} = \frac{-1}{2u} \cdot \frac{4}{\Sigma u^{-2}} = \frac{-2u^{-1}}{\Sigma u^{-2}}$$

$$\beta = \frac{-2v^{-1}}{\Sigma u^{-2}}, \quad \gamma = \frac{-2w^{-1}}{\Sigma u^{-2}}$$

$$\alpha^2 + \beta^2 + \gamma^2 = \frac{4}{(\Sigma u^{-2})^2}(\Sigma u^{-2}) = \frac{4}{\Sigma u^{-2}}$$

$\because \quad \Sigma u^{-2} = \dfrac{4}{\Sigma\alpha^2}$

$\therefore \quad \alpha = \dfrac{-2u^{-1}}{\Sigma u^{-2}} = \dfrac{-2u^{-1}}{4}\Sigma\alpha^2 = \dfrac{-1}{2}u^{-1}\Sigma\alpha^2.$

$\Rightarrow \quad u^{-1} = \dfrac{-2\alpha}{\Sigma\alpha^2}$ or $u = \dfrac{\Sigma\alpha^2}{-2\alpha}$

Similarly, $v = \dfrac{\Sigma\alpha^2}{-2\beta}$, $w = \dfrac{\Sigma\alpha^2}{-2r}$.

Putting the values of u, v, w in (2), we get

$$\frac{1}{4}(\Sigma\alpha^2)^2(\Sigma\alpha^{-2}) = k^2.$$

$\because$ Locus of (α, β, γ) is

$$(x^2+y^2+z^2)^2(x^{-2}+y^{-2}+z^{-2}) = 4k^2.$$

Example 4. A sphere whose centre lies in the positive octant passes through the origin and cuts the planes $x = 0$, $y = 0$, $z = 0$ in circles of radii $a\sqrt{2}$, $b\sqrt{2}$, $c\sqrt{2}$ respectively Show that its equation is

$$x^2+y^2+z^2-2\sqrt{(b^2+c^2-a^2)}\;x-2\sqrt{(c^2+a^2-b^2)}\;y-2\sqrt{(a^2+b^2-c^2)}\;z=0.$$

Solution : Suppose the equation of the sphere is

$$x^2+y^2+z^2-2ux-2vy-2wz+d=0 \quad \text{.... (1)}$$

$u, v, w > 0$.

(1) passes through the origin. $\therefore\ d = 0$.

Put $x = 0$ in (1), we get

$y^2+z^2-2vy-2wz=0.$ .... (2)

This is a circle in $yz-$ plane.

Its radius $= \sqrt{v^2+w^2}$

$\therefore \quad \sqrt{v^2+w^2} = a\sqrt{2}$

or $\quad v^2+w^2 = 2a^2$ .... (3)

Again, put $y = 0$ in (1) to get

$$x^2+z^2-2ux-2wz=0.$$

This is a circle in XZ–plane

$\therefore \quad u^2+w^2 = 2b^2$ .... (4)

Similarly $\quad u^2+v^2 = 2c^2$ .... (5)

Adding (3), (4) and (5), we get

$$2(u^2+v^2+w^2) = 2(a^2+b^2+c^2)$$

$\therefore \quad u^2+v^2+w^2 = a^2+b^2+c^2$ .....(6)

(3) & (6) $\Rightarrow u^2 = b^2+c^2-a^2 \Rightarrow u = \sqrt{b^2+c^2-a^2}$.

Similarly $\quad v = \sqrt{b^2+c^2-a^2} \Rightarrow w = \sqrt{a^2+b^2-c^2}$.

The required equation of the sphere then, from (1), is

$$x^2+y^2+z^2-2\sqrt{(b^2+c^2-a^2)}\;x-2\sqrt{(c^2+a^2-b^2)}\;y-2\sqrt{(a^2+b^2-c^2)}\;z=0.$$

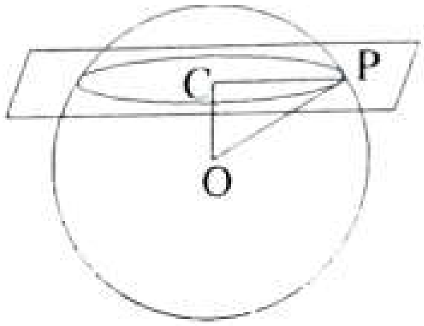
Questions for Practice:

1. Find the centre and radius of each of the following spheres:
   (i) $x^2 + y^2 + z^2 - 2x + 5z - 5 = 0$,
   (ii) $6(x^2 + y^2 + z^2) - 8x + 10y - 18z - 13 = 0$.
2. Find the equation of the sphere having its centre on the line
   $$5y + 2z = 0,\ 2x - 3y = 0$$
   and passing through the two points $(0, -2, -4)$, $(2, -1, -1)$.
3. Find the equation of the sphere which passes through the points $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, and has its radius as small as possible.
4. Find the equation of the sphere through the four points
   $$(0,0,0),\ (a, 0, 0),\ (0, b, 0),\ (0,0,c).$$
   Find its centre and radius also.
5. A sphere of constant radius 2k passes through the origin and meets the axes in A, B, C. Show that the locus of the centroid of the tetrahedron O ABC is the sphere
   $$x^2 + y^2 + z^2 = k^2.$$
6. Show that the locus of the centres of the spheres which pass through the points $(1,0,0)$, $(0, 2, 0)$ and $(2,3,4)$ is a straight line; find its equations.
7. Find the equation of the sphere described on the join of $(1, 2, 3)$ and $(0, 4, -1)$ as one of its diameters.

LECTURE NO.:- 12

INTERSECTION OF A SPHERE AND A PLANE

To show that the plane section of a sphere is a circle.



Let O be the centre of the sphere and let P be any point on the plane section. Let OC be the perpendicular form O to the plane: C is the foot of the perpendicular. Clearly $OC \perp CP$. From the right angled triangle OCP.

$$CP^2 = OP^2 - OC^2$$

Since O and C are fixed points, OC is a constant quantity and so is OP, the radius of the sphere. This yields therefore that CP is also constant and since C is a fixed point, the locus of P consequently is a circle. The centre of this circle is the foot of the perpendicular from the centre of the sphere to the given plane. The radius of the section is given by (1) above.

When the plane passes through the centre of the sphere, the section is called a **great circle.** The centre and radius of a great circle are the same as those of the sphere.

EQUATIONS OF A CIRCLE

A circle is the intersection of a sphere and a plane. Therefore a circle can be represented by two equations, one being of a sphere and the other being of a plane.

The two equations

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$$

and $\quad a'x + b'y + c'z + d' = 0$

taken together represent a circle.

A circle is also respresented by means of any two spheres passing through it. Let

$$S_1 \equiv x^2 + y^2 + z^2 + 2u_1 x + 2v_1 y + 2w_1 z + d_1 = 0,$$
$$S_2 \equiv x^2 + y^2 + z^2 + 2u_2 x + 2v_2 y + 2w_2 z + d_2 = 0$$

be two spheres; their points of intersection lie on the plane

$$S_1 - S_2 \equiv 2(u_1 - u_2)x + 2(v_1 - v_2)y + 2(w_1 - w_1)z + (d_1 - d_1) = 0.$$

Since the plane section of a sphere is a circle, the points of intersection of the two spheres lie on a circle lying in the above plane. Such a circle can be represented by means of equations of two intersecting spheres.

SPHERES THROUGH A GIVEN CIRCLE

The equation $S + \lambda U = 0$ represents a family of spheres passing through the circle whose equations are .

$$S \equiv x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0,$$
$$U \equiv a'x + b'y + c'z + d' = 0.$$

The equation $S + \lambda S' = 0$ represents a family of spheres passing through the circle whose equations are

$$S \equiv x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$$
$$S' \equiv x^2 + y^2 + z^2 + 2u'x + 2v'y + 2w'z + d' = 0.$$

Since the plane

$$S - S' \equiv 2(u - u')x + 2(v - v')y + 2(w - w')z + d - d' = 0$$

represents the plane of intersection of the spheres $S \equiv 0$, $S' \equiv 0$; the equation

$$S + \lambda(S - S') = 0$$

also represents the equation of a sphere through the circle

$$S \equiv 0,\ S' \equiv 0.$$

Ex. 1. Find the centre and radius of the circle

$$x^2 + y^2 + z^2 - 2y - 4z = 11;\ x + 2y + 2z = 15.$$

Solution. The cenre of the circle P is the foot of the $\perp$ drawn from the centre C (0, 1, 2) of the sphere on the plane

$$x + 2y + 2z = 15. \qquad \text{.... (1)}$$

Equation of the line through C (0, 1, 2) and $\perp$ to plane (1) is

$$\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-2}{2} (= r), \text{ say} \qquad \text{.... (2)}$$

Any point on this line is $(r, 2r + 1, 2r + 2)$.

Let P be $(r, 2r + 1, 2r + 2)$. It lies on (1).

$\therefore \quad r + 2(2r + 1) + 2(2r + 2) = 15$

$\Rightarrow \quad 9r = 9 \Rightarrow r = 1.$

$\therefore \quad$ P is (1, 3, 4).

The radius of the sphere is $= \sqrt{1 + 4 + 11} = 4.$

It Q is any point on the sphere, the radius of the circle is PQ, where

$$PQ^2 = CQ^2 - CP^2$$
$$= (4)^2 - (1 + 4 + 4)$$
$$= 16 - (9) = 7 \Rightarrow PQ = \sqrt{7}.$$

Ex. 2. Find the equations of the circle circumcribing the triangle formed by the three points

$$(a, 0, 0),\ (0, b, 0),\ (0, 0, c).$$

Obtain also the coordinates of the centre of this circle.

Solution. To find the equations of the circle, we have to find the equation of the sphere through the 3 given points and the equation of the plane through these 3 points. We need a 4th point for determining the equation of a sphere. Let it be (0, 0, 0). We now determine the equation of the sphere passing through 4 points

$(0, 0, 0), (a, 0, 0), (0, b, 0), (0, 0, c)$.

Suppose the equation of the sphere is

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad \text{.... (1)}$$

Then $d = 0$.

Also, $a^2 + 2ua = 0 \Rightarrow u = -\frac{a}{2}$.

Similarly, $v = -\frac{b}{2}, w = -\frac{a}{2}$.

Equation of the sphere is

$$x^2 + y^2 + z^2 - ax - by - cz = 0 \quad \text{.... (2)}$$

Equation of the plane through $(a, 0, 0), (0, b, 0), (0, 0, c)$ is

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1. \quad \text{.....(3)}$$

Equations (2) and (3) determine the circle.

To find the centre of the circle given by (2) and (3), let P be the foot of the $\perp$ from the centre $C\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$ of the sphere on the plane (3).

Equations of CP are

$$\frac{x - a/2}{1/a} = \frac{y - b/2}{1/b} = \frac{z - c/2}{1/c} (= r), \text{ say} \quad \text{.... (4)}$$

Any point on (4) is

$$\left(\frac{a}{2} + \frac{r}{a}, \frac{b}{2} + \frac{r}{b}, \frac{c}{2} + \frac{r}{c}\right).$$

Let P be $\left(\frac{a}{2} + \frac{r}{a}, \frac{b}{2} + \frac{r}{b}, \frac{c}{2} + \frac{r}{c}\right)$, for some r. P lies on (3).

$$\therefore \quad \left(\frac{1}{2} + \frac{r}{a^2}\right) + \left(\frac{1}{2} + \frac{r}{b^2}\right) + \left(\frac{1}{2} + \frac{r}{c^2}\right) = 1$$

$$r\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right) = -\frac{1}{2} \Rightarrow r = -\frac{1}{2}(a^{-2} + b^{-2} + c^{-2})^{-1}$$

$$\Rightarrow P\left[\frac{a}{2} - \frac{1}{2a(a^{-2}+b^{-2}+c^{-2})}, \frac{b}{2} - \frac{1}{2b(a^{-2}+b^{-2}+c^{-2})}, \frac{b}{2} - \frac{1}{2b(a^{-2}+b^{-2}+c^{-2})}\right].$$

Now $\frac{a}{2} - \frac{1}{2a(a^{-2}+b^{-2}+c^{-2})} = \frac{a^2(a^{-2}+b^{-2}+c^{-2})-1}{2a(a^{-2}+b^{-2}+c^{-2})} = \frac{a}{2}\frac{(b^{-2} + c^{-2})}{(\sum a^{-2})}$.

$$\Rightarrow P\left[\frac{a(b^{-2} + c^{-2})}{2\sum a^{-2}}, \frac{b(c^{-2} + a^{-2})}{2\sum a^{-2}}, \frac{c(a^{-2} + b^{-2})}{2\sum a^{-2}}\right].$$

This is the centre of the circle given by the equations (2) and (3).

Ex. 3. Find the equation of the sphere through the circle

$$x^2 + y^2 + z^2 + 2x + 3y + 6 = 0, x - 2y + 4z - 9 = 0$$

and (passing through) the centre of the sphere $x^2 + y^2 + z^2 - 2x + 4y - 6z + 5 = 0$.

Solution. Let the equation of the sphere through the given circle is

$$x^2 + y^2 + z^2 + 2x + 3y + 6 + \lambda(x - 2y + 4z - 9) = 0. \qquad \text{... (1)}$$

Given sphere is $x^2 + y^2 + z^2 - 2x + 4y - 6z + 5 = 0.$ ... (2)

Its centre is $(1, -2, 3)$.

If sphere (1) passes through the centre of the sphere (2), then

$$1 + 4 + 9 + 2 - 6 + 6 + \lambda(1 + 4 + 12 - 9) = 0$$

$\Rightarrow$ $16 + 8\lambda = 0 \Rightarrow \lambda = -2.$

Putting the value of λ in equation (1), we have

$$x^2 + y^2 + z^2 + 2x + 3y + 6 - 2(x - 2y + 4z - 9) = 0$$

$\Rightarrow$ $x^2 + y^2 + z^2 + 7y - 8z + 24 = 0.$

This is the required equation of the sphere.

Ex. 4. Find the centre and radius of the circle given by

$$x^2 + y^2 + z^2 - 2x - 4y - 6z - 2 = 0,\ x + 2y + 2z - 20 = 0.$$

Solution. Given circle is the intersection of

the sphere $x^2 + y^2 + z^2 - 2x - 4y - 6z - 2 = 0$ ... (1)

and the plane $x + 2y + 2z - 20 = 0.$ ... (2)

Let C_1 and C_2 be the centres of the sphere (1) and the given circle and r_1 and r_2 be the radii of the sphere (1) and the given circle. So from (1), we have

$C_1(-u, -v, -w) = C_1(1, 2, 3)$

and $r_1 = \sqrt{u^2 + v^2 + w^2 - d} = \sqrt{1 + 4 + 9 + 2} = 4.$

C_2 is the foot of the perpendicular from C_1 on the plane (2). So C_1C_2 is the normal to the plane (2), whose d.r.'s are 1, 2, 2.

Hence the equations of the line C_1C_2 are

$$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{2} = r \text{ (say)}.$$

Any point on the line C_1C_2 is $(r + 1, 2r + 2, 2r + 3)$. If this point is C_2, then it lies on the plane (2) so that

$(r + 1) + 2(2r + 2) + 2(2r + 3) - 20 = 0$, which gives $r = 1$.

So the coordinates of C_2 are obtained as $(2, 4, 5)$ and

length of $C_1C_2 = \sqrt{(2-1)^2 + (4-2)^2 + (5-3)^2} = 3.$

Now $r_1^2 = r_2^2 + (C_1C_2)^2$ gives $r_2^2 = r_1^2 - (C_1C_2)^2 = 16 - 9 = 7.$

So $r_2 = \sqrt{7}.$

Hence the centre and radius of the given circle are (2, 4, 5) and $\sqrt{7}$. Ans.

Example 5. Find the equation of the sphere having the circle

$$x^2+y^2+z^2+10y-4z-8=0,\quad x+y+z=3$$

as a great circle.

Solution: The equation of any sphere passing through the given circle is of the form

$$x^2+y^2+z^2+10y-4z-8+\lambda(x+y+z-3)=0$$

$$\text{i.e., } x^2+y^2+z^2+\lambda x+(\lambda+10)+(\lambda-4)z-(3\lambda+8)=0. \qquad \ldots(1)$$

If the given circle is a great circle of sphere (1), centre of sphere (1) should be on the plane $x+y+z-3=0$ in which the given circle lies.

$\therefore$ Centre of sphere (1) is $\left(-\frac{\lambda}{2}, \frac{-(\lambda+10)}{2}, \frac{-(\lambda-4)}{2}\right)$.

$$\therefore -\frac{\lambda}{2}-\frac{1}{2}(\lambda+10)-\frac{1}{2}(\lambda-4)=3.$$

It gives $\lambda=-4$.

$\therefore$ Equation of the required sphere is $x^2+y^2+z^2-4x+6y-8z+4=0$.

LECTURE NO.:- 13

TANGENT PLANE

EQUATION OF A TANGENT PLANE TO A SPHERE

To find the equation of the tangent plane at any point P (x_1, y_1, z_1) of the sphere

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0. \quad \text{....(1)}$$

The point P (x_1, y_1, z_1) lies on the sphere.

$$\therefore x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1 + d = 0. \quad \text{....(2)}$$

The line joining the centre of a sphere to any point on it is perpendicular to the tangent plane at that point. The direction ratios of the normal to the tangent plane at $P(x_1, y_1, z_1)$ are therefore the same as the direction ratios of the line CP, where C is $(-u, -v, -w)$, the centre of the sphere, and as such are

$$x_1 + u, y_1 + v, z_1 + w.$$

The equation of the tangent plane at P is, therefore,

$$(x - x_1)(x_1 + u) + (y - y_1)(y_1 + v) + (z - z_1)(z_1 + w) = 0$$

or, $x(x_1 + u) + y(y_1 + v) + z(z_1 + w) - ux_1 - vy_1 - wz_1$
$= x_1^2 + y_1^2 + z_1^2,$

or, $x(x_1 + u) + y(y_1 + v) + z(z_1 + w) - ux_1 - vy_1 - wz_1$
$= -2ux_1 - 2vy_1 - 2wz_1 - d.$ using (2)

or $x(x_1 + u) + y(y_1 + v) + z(z_1 + w) + ux_1 + vy_1 + wz_1 + d = 0.$

Or $xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0.$ ... (3)

Here point P (x_1, y_1, z_1) is called the point of contact of the sphere (1) and the plane (3).

CONDITION OF TANGENCY

To find the condition that the plane

$$Ax + By + Cz + D = 0$$

may touch the sphere

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0.$$

The given plane will touch the sphere if the perpendicular distance of the plane from the centre of the sphere is equal to the radius of the sphere.

The required condition of tangency is, therefore,

$$\pm \frac{-uA - vB - wC + D}{\sqrt{(A^2 + B^2 + C^2)}} = \sqrt{(u^2 + v^2 + w^2 - d)}.$$

On squaring both sides, we get

$$(uA + vB + wC - D)^2 = (u^2 + v^2 + w^2 - d)(A^2 + B^2 + C^2).$$

Ex. 1. Find the coordinates of the points on the sphere

$$x^2 + y^2 + z^2 - 4x + 2y - 4 = 0,$$

the tangent planes at which are parallel to the plane

$$2x - y + 2z = 1. \quad \textbf{[Raj., 2001]}$$

Solution Equation of the plane parallel to the plane
$2x - y + 2z = 1$,
is $2x - y + 2z = \lambda$. ....(1)
This is a tangent plane to the given sphere :
$x^2 + y^2 + z^2 - 4x + 2y - 4 = 0$. ....(2)
Its centre is $(2, -1, 0)$ & radius $= \sqrt{4 + 1 + 4} = 3$.
Using the fail that $\perp$ distance of the tangent plane (1) from $(2, -1, 0)$ $= 3$.

$$\Rightarrow \quad \frac{|2(2) - (-1) + 2(0) - \lambda|}{\sqrt{4 + 1 + 4}} = 3.$$

$\Rightarrow \quad 5 - \lambda = \pm 9 \Rightarrow \lambda = -4, 14$.

$\therefore$ Equation of the tangent planes are

$2x - y + 2z + 4 = 0$ ....(3)

and $2x - y + 2z = 14$ ....(4)

Let (α, β, γ) be the point of contact of the tangent plane (3).
The equation of the tangent plane at (α, β, γ) to the given sphere is

$\alpha x + \beta y + \gamma z - 2(x + \alpha) + (y + \beta) = 4$

$\Rightarrow \quad (\alpha - 2)x + (\beta - 1)y + \gamma z = 2\alpha - \beta + 4$ ....(5)

(3) and (5) are same

$$\therefore \quad \frac{\alpha - 2}{2} = \frac{\beta + 1}{-1} = \frac{\gamma}{2} = \frac{2\alpha - \beta + 4}{-4}$$

These give, $-\alpha + 2 = 2\beta + 2 \Rightarrow \alpha = -2\beta$;
$2\beta + 2 = -\gamma \Rightarrow \gamma = -2\beta - 2$;
$-2\gamma = 2\alpha - \beta + 4$

$\Rightarrow \quad 4\beta + 4 = 4\beta - \beta + 4$

$\Rightarrow \quad 9\beta = 0 \Rightarrow \beta = 0$.

$\alpha = 0, \quad \gamma = -2$.

One point of contact is $(0, 0, -2)$.
Similarly, (4) & (5) are same.

$$\therefore \quad \frac{\alpha - 2}{2} = \frac{\beta + 1}{-1} = \frac{\gamma}{2} = \frac{2\alpha - \beta + 4}{14}$$

$\Rightarrow \quad \alpha = -2\beta, \gamma = -2\beta - 2, 7\gamma = 2\alpha - \beta + 4$

$\Rightarrow \quad 9\beta = -18 \Rightarrow \beta = -2$

$\alpha = 4, \gamma = +2 \quad \therefore$ Another point of contact is $(4, -2, 2)$.

Ex. 2. Tangent plane at any point of the sphere $x^2 + y^2 + z^2 = r^2$ meets the coordinate axes at A, B, C. Show that the locus of the point of intersection of planes drawn parallel to the coordinate planes through A, B, C is the surface

$x^{-2} + y^{-2} + z^{-2} = r^{-2}$ **[Raj., 2003]**

Solution. Let (α, β, γ) be any point on the sphere
$x^2 + y^2 + z^2 = r^2$ ....(1)

$\Rightarrow \quad \alpha^2 + \beta^2 + \gamma^2 = r^2$ ....(2)

Tangent plane at (α, β, γ) to (1) is

$\alpha x + \beta y + \gamma z = r^2$ ...(3)

(3) meets the axes in A, B, C.

$\therefore \quad A\left[\frac{r^2}{\alpha}, 0, 0\right], B\left[0, \frac{r^2}{\beta}, 0\right], C\left[0, 0, \frac{r^2}{\gamma}\right].$

Equation of a plane parallel to yz–plane is

$$x = \lambda_1.$$

It passes thrugh A. $\therefore \lambda_1 = \frac{r^2}{\alpha}$.

$\Rightarrow$ Equation of the plane parallel to yz–plane and through A is $x = \frac{r^2}{\alpha}$.

Similarly, the other two planes are $y = \frac{r^2}{\beta}$, $z = \frac{r^2}{\gamma}$.

The point of intersection of these three planes is

$$\left(\frac{r^2}{\alpha}, \frac{r^2}{\beta}, \frac{r^2}{\gamma}\right).$$

To find its locus, suppose this point is (λ, μ, ν).

$$\therefore \quad \lambda = \frac{r^2}{\alpha}, \mu = \frac{r^2}{\beta}, \nu = \frac{r^2}{\gamma}.$$

Putting the value of α, β and γ from above equations in equation (2), the required locus is obtained.

Example 3. Find the equations of the tangent planes to the sphere $x^2 + y^2 + z^2 = 9$, which pass through the line $x + y = 6$, $x - 2z = 3$.

Solution. Equation of any plane passing through the given line $x + y = 6$, $x - 2z = 3$ is

$$(x + y - 6) + \lambda(x - 2z - 3) = 0$$

$\Rightarrow \quad (1 + \lambda)x + y - 2\lambda z - 6 - 3\lambda = 0.$ ... (1)

Given sphere is $x^2 + y^2 + z^2 = 3^2$. ... (2)

The centre of the sphere (2) is (0, 0, 0) and its radius is 3.

If plane (1) is the tangent plane to the sphere (2), then the perpendicular distance of the centre (0, 0, 0) from plane (1) is equal to the radius of the sphere (2)

i.e. $\left|\frac{0+0-0-6-3\lambda}{\sqrt{(1+\lambda)^2+1^2+(-2\lambda)^2}}\right| = 3$ i.e. $\frac{3|2+\lambda|}{\sqrt{1+2\lambda+\lambda^2+1+4\lambda^2}} = 3$

i.e. $|2+\lambda| = \sqrt{5\lambda^2 + 2\lambda + 2}$ i.e. $(2+\lambda)^2 = 5\lambda^2 + 2\lambda + 2$

i.e. $4\lambda^2 - 2\lambda - 2 = 0$ i.e. $2\lambda^2 - \lambda - 1 = 0$

i.e. $(2\lambda + 1)(\lambda - 1) = 0$ i.e. $\lambda = -\frac{1}{2}, 1.$

Putting these values of λ in equation (1), we get

$(1 - 1/2)x + y - 2(-1/2)z - 6 - 3(-1/2) = 0$, $(1 + 1)x + y - 2(1)z - 6 - 3(1) = 0$

i.e. $(1/2)x + y + z - 6 + 3/2 = 0$, $2x + y - 2z - 6 - 3 = 0$

i.e. $x + 2y + 2z - 9 = 0$, $2x + y - 2z - 9 = 0$.

These are the required equations of the tangent planes.

Example 4. Find the equations of tangent planes of the sphere $x^2+y^2+z^2-4x-4y-4z+10=0$, which are parallel to the plane $x-z=0$.

Solution. Let (x_1,y_1,z_1) be the point on the sphere at which the tangent plane is drawn. The equation of the tangent plane at (x_1, y_1, z_1) is $xx_1+yy_1+zz_1-2(x+x_1)-2(y+y_1)-2(z+z_1)+10=0$

i.e., $(x_1-2)x+(y_1-2)y+(z_1-2)z-2x_1-2y_1-2z_1+10=0.$ ...(1)

This plane is parallel to $x-z=0$.

$$\therefore \frac{x_1-2}{1}=\frac{y_1-2}{0}=\frac{z_1-2}{-1}=k \ (say)$$

$\Rightarrow x_1=k+2, \quad y_1=2, \ z_1=-k+2.$ ...(2)

Since (x_1,y_1,z_1) lies on the sphere, we have

$$(k+2)^2+2^2+(-k+2)^2-4(k+2)-8-4(-k+2)+10=0.$$

$\therefore k^2-1=0$, i.e. $k=\pm 1$.

from (2), the points are (3,2,1) and (1,2,3).

$\therefore$ The equation of the tangent planes are $x-z-2=0$ and $-x+z-2=0$.

Example 5. Find the equation of the sphere which pass through the circle

$$x^2+y^2+z^2-2x+2y+4z-3=0, \quad 2x+y+z-4=0$$

and touch the plane $3x+4y-14=0$.

Solution. Given circle is the intersection of the sphere and the plane, i.e.

$S: x^2+y^2+z^2-2x+2y+4z-3=0$ ... (1)

and $P: 2x+y+z-4=0$. ...(2)

Then $S+\lambda P=0$ represents a sphere passing through the circle determined by (1) & (2).

$\therefore S+\lambda P=(x^2+y^2+z^2-2x+2y+4z-3)+\lambda(2x+y+z-4)=0$

i.e., $x^2+y^2+z^2-2x(1-\lambda)+y(2+\lambda)+z(4+\lambda)-(3+4\lambda)=0$. ... (3)

Centre of the sphere (3) is

$$\left(1-\lambda, \frac{-(2+\lambda)}{2}, \frac{-(4+\lambda)}{2}\right) \text{ and}$$

$$\text{radius} = \sqrt{(1-\lambda)^2+\left(\frac{2+\lambda}{2}\right)^2+\left(\frac{4+\lambda}{2}\right)^2+(3+4\lambda)}.$$

Since the sphere touches the plane $3x+4y-14=0$, the perpendicular distance from the centre of the sphere to this plane is equal to the radius of the sphere.

$$\therefore \quad \frac{3(1-\lambda)-2(2+\lambda)-14}{\sqrt{3^2+4^2}} = \sqrt{(1-\lambda)^2+\left(\frac{2+\lambda}{2}\right)^2+\left(\frac{4+\lambda}{2}\right)^2+(3+4\lambda)}$$

i.e. $2\lambda^2 - 4\lambda = 0$, i.e. $\lambda(\lambda-2) = 0$, i.e. $\lambda = 0, \lambda = 2$.

Using these values in (3), we get

$$x^2+y^2+z^2-2x+2y+4z-3=0$$

and $x^2+y^2+z^2+2x+4y+6z-11=0$.

Example 6. Show that the plane $2x - 2y + z + 12 = 0$ touches the sphere $x^2 + y^2 + z^2 - 2x - 4y + 2z = 3$. Find also the point of contact.

Solution. Given sphere is $x^2 + y^2 + z^2 - 2x - 4y + 2z = 3$. ... (1)

Centre of the sphere (1) is (1,2,-1)

Radius is $\sqrt{1+4+1+3}=3$

Given plane is $2x - 2y + z + 12 = 0$. ... (2)

Length of perpendicular from (1,2,-1) to the plane (2) is

$$=\frac{|2-4-1+12|}{\sqrt{4+4+1}}$$

$=3$

i.e., Length of perpendicular from (1,2,-1) to the plane (2) = radius of the sphere

$\therefore$ The plane (2) touches the sphere (1).

Let P be the point of contact. Then CP is normal to the plane (2)

i.e., Normal line of the plane (2) and the line CP are parallel.

So their d.r.'s are proportional.

Hence the d.r.'s of the line CP are 2, –2, 1.

Also the line CP passes through the point C(1, 2, –1).

Therefore the equations of the line CP are

$$\frac{x-1}{2}=\frac{y-2}{-2}=\frac{z+1}{1}=r$$

Any point on the line CP is $(2r + 1, -2r + 2, r - 1)$.

If this point lies on the plane (2), it will be the point of contact P. Then

$2(2r + 1) - 2(-2r + 2) + (r - 1) + 12 = 0$, which gives $r = -1$.

Hence the point of contact is P(–1, 4, –2).

LECTURE NO.:- 14

INTERSECTION OF TWO SPHERES

Let us consider two spheres given by

$S_1 \equiv x^2 + y^2 + z^2 + 2u_1x + 2v_1y + 2w_1z + d_1 = 0$, and

$S_2 \equiv x^2 + y^2 + z^2 + 2u_2x + 2v_2y + 2w_2z + d_2 = 0$.

Then each point that satisfies $S_1 = 0$ as well as $S_2 = 0$ will also satisfy the equation $S_1 - S_2 = 0$, that is,

$$2(u_1 - u_2)x + 2(v_1 - v_2)y + 2(w_1 - w_2)z + d_1 - d_2 = 0.$$

But, we know that this is a plane. Thus, the points of intersection of the spheres $S_1 = 0$ and $S_2 = 0$ are the same as those of any one of the spheres and the above plane. Since the intersection of a plane and a sphere is a circle, the intersection of $S_1 = 0$ and $S_2 = 0$ is a circle.

If $S_1 = 0$ and $S_2 = 0$ are any two spheres, which do not necessarily intersect, the. $S_1 - S_2 = 0$ is a plane, and is called the radical plane of the spheres.

ANGLE OF INTERSECTION OF TWO SPHERES

The angle of intersection of two spheres is defined to be the angle between the tangent planes to these spheres at a point of contact. As the radii of the spheres at this common point are normal to the tangent planes so this angle is also equal to the angle between the radii of the spheres at their common point of intersection. If the angle of intersection of two spheres is a right angle, the spheres are said to be orthogonal.

The angle of intersection of two spheres is defined to be the angle between the tangent planes to these spheres at a point of contact.

Let the spheres be $S_1 = 0$ and $S_2 = 0$, where

$S_i \equiv x^2 + y^2 + z^2 + 2u_ix + 2v_iy + 2w_iz + d_i = 0$ and $i = 1, 2$.

Then their radii are $r_1 = \sqrt{u_1^2 + v_1^2 + w_1^2 - d_1}$ and $r_2 = \sqrt{u_2^2 + v_2^2 + w_2^2 - d_2}$ respectively.

Let C_1 and C_2 be the centres of the spheres and P be the point of intersection of spheres.

Then the angle between the spheres is the angle between the tangent planes at P to each of the spheres. This, in turn, is the angle between the normals to these planes, which are PC_1 and PC_2. Thus, if θ is the required angle, then from triangle PC_1C_2,

$$2PC_1 \cdot PC_2 \cdot \cos\theta = PC_1^2 + PC_2^2 - C_1C_2^2.$$

$$\therefore \cos\theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}$$

Thus, in particular, the spheres will be orthogonal iff $r_1^2 + r_2^2 = d^2$

$$\Rightarrow (u_1^2 + v_1^2 + w_1^2 - d_1) + (u_2^2 + v_2^2 + w_2^2 - d_2) = (-u_1 + u_2)^2 + (-v_1 + v_2)^2 + (-w_1 + w_2)^2$$

$$\Rightarrow 2u_1u_2 + 2v_1v_2 + 2w_1w_2 = d_1 + d_2$$

Thus, the spheres $S_1 = 0$ and $S_2 = 0$ intersect at 90^o, iff above condition is satisfied.

TOUCHING SPHERES

To find the point of contact if the two given spheres touch internally or externally

Let S_1 be the sphere with centre $C_1(-u_1,-v_1,-w_1)$ and radius r_1 .

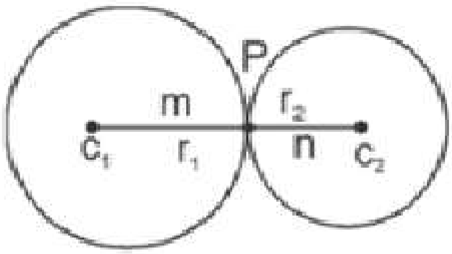
Let S_2 be the sphere with centre $C_2(-u_2,-v_2,-w_2)$ and radius r_2 .

Let P be the point of contact.

Case I

Two spheres Touch externally

The point of contact is the point which divides the line joining the two points C_1 and C_2 in the ratio m : n internally.



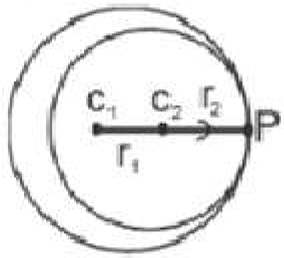
Hence the coordinates of the point of contact are

$$P\left(\frac{m(-u_2)+n(-u_1)}{m+n},\frac{m(-v_2)+n(-v_1)}{m+n},\frac{m(-w_2)+n(-w_1)}{m+n}\right).$$

Case II

Two spheres touch internally

The point of contact is the point which divides the line joining the two points C_1 and C_2 in the ratio m: n externally.



Hence the co-ordinates of the point of contact are

$$P\left(\frac{m(-u_2)-n(-u_1)}{m-n},\frac{m(-v_2)-n(-v_1)}{m-n},\frac{m(-w_2)-n(-w_1)}{m-n}\right).$$

Note:

(1) Two spheres S_1 and S_2 whose radii are r_1 and r_2 touch externally, if the distance between their centres is equal to the sum of their radii i.e. $C_1C_2 = r_1 + r_2$.

(2) Two spheres S_1 and S_2 whose radii are r_1 and r_2 touch internally, if the distance between their centres is equal to the difference of the radii i.e. $C_1C_2 = r_1 - r_2$.

Example 1. Prove that the two spheres

$$x^2+y^2+z^2-2x+4y-4z=0 \text{ and } x^2+y^2+z^2+10x+2z+10=0$$

touch each other and find the coordinates of the point of contact.

Solution: Let $S_1 : x^2+y^2+z^2-2x+4y-4z=0$ and $S_2 : x^2+y^2+z^2+10x++2z+10=0$.

The centre of S_1 is $C_1(1,-2,2)$ and the centre of S_2 is $C_2(-5,0,-1)$.

Radius of S_1 is $r_1=\sqrt{1+4+4}=3$.

Radius of S_2 is $r_2=\sqrt{25+1-10}=4$.

Distance between C_1 and C_2 is $C_1C_2=\sqrt{(1+5)^2+(-2+0)^2+(2+1)^2}=7$.

$\therefore\ r_1+r_2=3+4=7=C_1C_2$.

Hence the two given spheres touch externally.

To find their point of contact:

The point of contact is the point which divides the line joining the two points $C_1(1,-2,2)$ and $C_2(-5,0,-1)$ in the ratio 3 : 4 internally. Hence the coordinates of the point of contact are

$$P\left(\frac{3(-5)+4(1)}{3+4},\frac{3(0)+4(-2)}{3+4},\frac{3(-1)+4(2)}{3+4}\right)$$

(ie) $P\left(\frac{-11}{7},\frac{-8}{7},\frac{5}{7}\right)$.

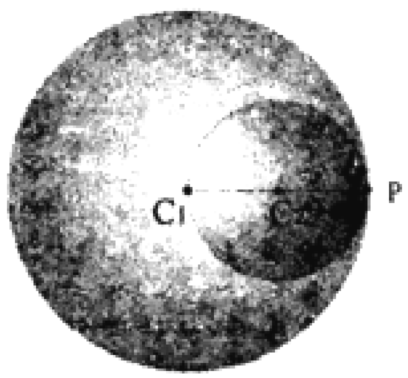
Example 2. Find the angle between spheres $x^2+y^2+z^2=4$ and $x^2+y^2+z^2-2x=0$.

Solution : Here $u_1=0, v_1=0, w_1=0, d_1=-4, u_2=-1, v_2=0, w_2=0, d_2=0$.

Thus, the centres of the two spheres are (0, 0, 0) and (1, 0, 0), their radii are 2 and 1, respectively, and the distance between their centres is 1.

Therefore, by (12), the angle between the two spheres is

$$\cos^{-1}\left(\frac{2^2+1^1-1^2}{2(2)(1)}\right)=\cos^{-1}(1)=0.$$



In above figure, spheres intersect in only one point P and the *x*-axis is the normal from the centres of the spheres to both tangent planes.

Example 3. Show that the spheres $x^2 + y^2 + z^2 - 2x - 3 = 0$ and $x^2+y^2+z^2+6x+6y+9=0$ **touch externally.**

Sol. The centre and radius of the sphere $x^2+y^2+z^2-2x-3=0$ is $(1, 1, 0)$ and $\sqrt{(1+3)}$ *i.e.* 2.

The centre and radius of the sphere $x^2+y^2+z^2+6x+6y+9=0$ is $(-3, -3, 0)$ and $\sqrt{(9+9-9)}$ *i.e.* 3

Now the distance between the centres

$$=\sqrt{[(1+3)^2+(0+3)^2+(0-0)^2]}=\sqrt{(16+9)}=5$$

And the sum of radii of the spheres $=2+3=5$

$\therefore$ The sum of the radii $=$ distance between the centres so the two spheres touch externally.

Example 4. Find the equation of the sphere through the circle

$$x^2 + y^2 + z^2 - x + 9y - 5z - 5 = 0, \quad x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$$

as a great circle.

Solution. Given spheres are

$S_1 : x^2 + y^2 + z^2 - x + 9y - 5z - 5 = 0$

and $S_2 : x^2 + y^2 + z^2 + 10y - 4z - 8 = 0$.

$S_1 - S_2 = 0$ represents a plane, which is $x + y + z - 3 = 0$. ... (1)

So $S_1 = 0, S_1 - S_2 = 0$ determines a circle.

Hence the equation of the sphere passing through above circle is

$S_1 + \lambda (S_1 - S_2) = 0$

i.e. $(x^2 + y^2 + z^2 - x + 9y - 5z - 5) + \lambda (x+y+z-3) = 0$

i.e. $x^2 + y^2 + z^2 + (\lambda - 1)x + (9 + \lambda) y + (\lambda - 5) z - 5 - 3\lambda = 0$. ... (2)

Its centre is $\left(\frac{1-\lambda}{2}, -\left(\frac{9+\lambda}{2}\right), \left(\frac{5-\lambda}{2}\right)\right)$.

Since the cross section is a great circle, above centre lies on the plane (1).

$$\therefore \quad \frac{1-\lambda}{2} - \left(\frac{9+\lambda}{2}\right) + \frac{5-\lambda}{2} - 3 = 0.$$

It gives $\lambda = -3$.

Putting this value of λ in equation (2), we get

$x^2 + y^2 + z^2 - 4x + 6y - 8z + 4 = 0$.

This is the required equation of the sphere.

LECTURE NO.:- 15

ORTHOGONALITY OF TWO SPHERES

Orthogonal spheres

Two spheres are said to cut each other orthogonally if the tangent planes at a point of intersection are at right angles.

If two spheres cut orthogonally at P, their radii through P, being perpendicular to the tangent planes at P, will also be at right angles.

CONDITION FOR THE ORTHOGONALITY OF TWO SPHERES

Let us consider the two spheres

$$x^2+y^2+z^2+2u_1x+2v_1y+2w_1z+d_1=0 \qquad \text{\_\_\_\_(1)}$$

and $$x^2+y^2+z^2+2u_2x+2v_2y+2w_2z+d_2=0. \qquad \text{\_\_\_\_(2)}$$

Let (1) and (2) cut orthogonally at P.

For (1), centre C_1 is $(-u_1,-v_1,-w_1)$

and radius $r_1=\sqrt{u_1^2+v_1^2+w_1^2-d_1}$. ____(3)

For (2), centre C_2 is $(-u_2,-v_2,-w_2)$

and radius $r_2=\sqrt{u_2^2+v_2^2+w_2^2-d_2}$. ____(4)

Also $C_1C_2=\sqrt{(u_1-u_2)^2+(v_1-v_2)^2+(w_1-w_2)^2}$. ____(5)

If spheres (1) and (2) are orthogonal, then

$(C_1C_2)^2=r_1^2+r_2^2$. -------(6)

Putting the values from (3), (4) and (5) in (6), we get

$$\boxed{2u_1u_2+2v_1v_2+2w_1w_2=d_1+d_2}$$

This is the condition for the orthogonality of two spheres.

Example 1. Find the equation of that sphere through the circle

$$x^2+y^2+z^2-2x+3y-4z+6=0,\ 3x-4y+5z-15=0,$$

which cuts the sphere $x^2+y^2+z^2+2x+4y-6z+11=0$ orthogonally.

Solution. Given circle is

$x^2+y^2+z^2-2x+3y-4z+6=0,\ 3x-4y+5z-15=0.$ ... (1)

The equation of any sphere passing through the given circle (1) is $S+\lambda P=0$

i.e. $x^2+y^2+z^2-2x+3y-4z+6=0+\lambda(3x-4y+5z-15)=0$

or $x^2+y^2+z^2+(3\lambda-2)x+(3-4\lambda)y+(5\lambda-4)z+(6-15\lambda)=0.$ ... (2)

Given sphere is $x^2+y^2+z^2-3x+5y-7z-6=0.$ ... (3)

If spheres (2) and (3) cut orthogonally, then

$$2u_1u_2 + 2v_1v_2 + 2w_1w_2 = d_1 + d_2$$

i.e. $(3\lambda - 2)(1) + (3 - 4\lambda)(2) + (5\lambda - 4)(-3) = (6 - 15\lambda) + 11$
i.e. $(3\lambda - 2) + 2(3 - 4\lambda) - 3(5\lambda - 4) = 17 - 15\lambda$
i.e. $(3 - 8 - 15 + 15)\lambda = 2 - 6 - 12 + 17$ i.e. $-5\lambda = 1$ i.e. $\lambda = -1/5$.

Putting this value of λ in equation (2), we get
$x^2 + y^2 + z^2 + (-3/5 - 2)x + (3 + 4/5)y + (-1 - 4)z + (6 + 3) = 0$
i.e. $5(x^2 + y^2 + z^2) - 13x + 19y - 25z + 45 = 0$.
This is the required equation of the sphere. Ans.

Example 2. Find the equation of the sphere that passes through the circle
$$x^2 + y^2 + z^2 + x - 3y + 2z - 1 = 0,\ 2x + 5y - z + 7 = 0$$
and cuts orthogonally the sphere $x^2 + y^2 + z^2 - 3x + 5y - 7z - 6 = 0$.

Solution. The equation of any sphere passing through the given circle is
$$x^2 + y^2 + z^2 + x - 3y + 2z - 1 + \lambda(2x + 5y - z + 7) = 0$$
i.e., $x^2 + y^2 + z^2 + (1 + 2\lambda)x + (-3 + 5\lambda)y + (2 - \lambda)z - 1 + 7\lambda = 0$ (1)

Sphere (1) cuts the sphere $x^2 + y^2 + z^2 - 3x + 5y - 7z - 6 = 0$ orthogonally.

∴ $2uu_1 + 2vv_1 + 2ww_1 = d + d_1$

i.e., $2\left(\frac{1+2\lambda}{2}\right)\left(\frac{-3}{2}\right) + 2\left(\frac{-3+5\lambda}{2}\right)\left(\frac{5}{2}\right) + 2\left(\frac{2-\lambda}{2}\right)\left(\frac{-7}{2}\right) = -1 + 7\lambda - 6$

i.e., $-3 - 6\lambda - 15 + 25\lambda - 14 + 7\lambda = -14 + 14\lambda$

i.e., $12\lambda - 18 = 0$ i.e., $\lambda = 3/2$

∴ Equation of the required sphere is
$$x^2 + y^2 + z^2 + x - 3y + 2z - 1 + (3/2)(2x + 5y - z + 7) = 0$$
i.e., $2(x^2 + y^2 + z^2) + 8x + 9y + z + 19 = 0$

Example 3. Prove that the two spheres $x^2 + y^2 + z^2 + 6y + 2z + 8 = 0$ and $x^2 + y^2 + z^2 + 6x + 8y + 4z + 20 = 0$ intersect each other orthogonally.

Solution. Given spheres are
$x^2 + y^2 + z^2 + 6y + 2z + 8 = 0$
and $x^2 + y^2 + z^2 + 6x + 8y + 4z + 20 = 0$.

From the equations of above spheres, we have

$u_1 = 0,\ v_1 = 3,\ w_1 = 1,\ d_1 = 8;\ u_2 = 3,\ v_2 = 4,\ w_2 = 2,\ d_2 = 20$.

By the condition of orthogonality,

$2u_1u_2 + 2v_1v_2 + 2w_1w_2 - (d_1+d_2) = 0 + 24 + 4 - (8+20) = 0$.

Hence the two spheres intersect orthogonally.

Questions for practice:

1. Find the equation of the sphere that passes through the points $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ and cuts the sphere $x^2 + y^2 + z^2 - 2ax - 2by - 2cz = 0$ orthogonally.
2. Find the equation of the sphere, which touches the plane $3x + 2y - z + 2 = 0$ at the point $(1, -2, 1)$ and cuts the sphere $x^2 + y^2 + z^2 - 4x + 6y + 4 = 0$ orthogonally.

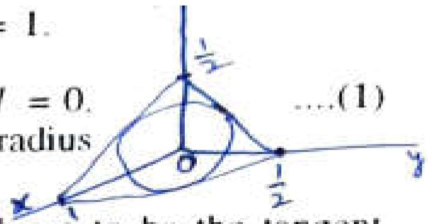
3. Two spheres of radii r_1 and r_2 cut orthogonally. Prove that the radius of the common circle is $r_1 r_2/\sqrt{r_1^2 + r_2^2}$.
4. Prove that the spheres that can be drawn through the origin and each set of points where the planes parallel to the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=0$ cut the coordinate axes, form a system of spheres which are cut orthogonally by the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$, if $au + bv + cw = 0$.
5. Find the equation of the sphere that passes through the two points (0, 3, 0) and (– 2, – 1, – 4) and cuts orthogonally the two spheres $x^2 + y^2 + z^2 + x - 3z + 4 = 0$ and $2(x^2 + y^2 + z^2) + x + 3y + 4 = 0$.
6. Find the equation of a sphere that passes through the circle $x^2 + y^2 + z^2 + 2x + y - 3z - 1 = 0, - x + 2y + 5z + 7 = 0$ and cuts the sphere $x^2 + y^2 + z^2 - 7x - 3y - 6 = 0$ orthogonally.
7. Find the equation of the sphere which cuts the following spheres orthogonally:
$x^2 + y^2 + z^2 = a^2 + b^2 + c^2$,
$x^2 + y^2 + z^2 + 2ax = a^2$,
$x^2 + y^2 + z^2 + 2by - 6 = b^2$,
$x^2 + y^2 + z^2 + 2cz = c^2$.
8. Show that every sphere through the circle $x^2 + y^2 - 2ax + r^2 = 0, z = 0$ cuts orthogonally every sphere through the circle $x^2 + z^2 = r^2, y = 0$.
9. Prove that the sphere which cuts two spheres $S_1 = 0$ and $S_2 = 0$ at right angles, will also cut $lS_1 + mS_2 = 0$ at right angles.

LECTURE NO.:- 16

MISCELLANEOUS PROBLEMS BASED ON SPHERE

Example 1. Find the equation of the sphere inscribed in a tetrahedron whose faces are

$$x = 0,\ y = 0,\ z = 0,\ x + 2y + 2z = 1.$$



Solution. Let the equation of the sphere be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0. \quad \text{....(1)}$$

The centre of this sphere is $(-u, -v, -w)$ and radius

$$= \sqrt{u^2 + v^2 + w^2 - d}\,.$$

The planes $x = 0,\ y = 0,\ z = 0,\ x + 2y + 2z = 1$ are to be the tangent planes to the sphere (1), for which

$$-u = -v = -w = \pm\frac{(-u - 2v - 2w - 1)}{3} = \sqrt{(u^2 + v^2 + w^2 - d)} \quad \text{....(2)}$$

From (2), we have $u = v = w$.

Also $\quad -u = +\dfrac{(-u - 2v - 2w - 1)}{3} \Rightarrow u = \dfrac{5u+1}{3} \Rightarrow u = -\dfrac{1}{2}$

and $\quad -u = -\dfrac{(-u - 2v - 2w - 1)}{3} \Rightarrow u = \dfrac{-5u - 1}{3} \Rightarrow u = -1/8.$

Again, $\quad -u = \sqrt{u^2 + v^2 + w^2 - d} \Rightarrow -u = \sqrt{3u^2 - d}$

$$\Rightarrow u^2 = 3u^2 - d \Rightarrow d = 2u^2.$$

When $u = -\dfrac{1}{2},\ d = \dfrac{1}{2}$; when $u = -\dfrac{1}{8},\ d = \dfrac{1}{32}$

Thus, the centre of the sphere is $\left(\dfrac{1}{8}, \dfrac{1}{8}, \dfrac{1}{8}\right)$, the other centre $\left(\dfrac{1}{2}, \dfrac{1}{2}, \dfrac{1}{2}\right)$ being the centre of the escribed sphere.

The equation of the inscribed sphere is

$$x^2 + y^2 + z^2 - \frac{1}{4}x - \frac{1}{4}y - \frac{1}{4}z + \frac{1}{32} = 0.$$

or $\quad 32(x^2 + y^2 + z^2) - 8(x + y + z) + 1 = 0.$

Example 2. Write the equation of the tangent plane at (1, 5, 7) to the sphere

$$(x - 2)^2 + (y - 3)^2 + (z - 4)^2 = 14.$$

Solution. The equation of the tangent plane to the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ at any point (x_1, y_1, z_1) is

$$xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0 \quad \textbf{(1)}$$

Given : $(x-2)^2 + (y-3)^2 + (z-4)^2 = 14$

$(x^2 - 4x + 4) + (y^2 - 6y + 9) + (z^2 - 8z + 16) = 14$

$x^2 + y^2 + z^2 - 4x - 6y - 8z + 29 - 14 = 0$

Here 2u = -4, 2v = -6, 2w = -8, d = 15

$x_1 = 1,\ y_1 = 5,\ z_1 = 7$

$(1) \Rightarrow x(1) + y(5) + z(7) + (-2)(x+1) + (-3)(y+5) + (-4)(z+7) + 15 = 0$

$x + 5y + 7z - 2x - 2 - 3y - 15 - 4z - 28 + 15 = 0$

$-x + 2y + 3z - 30 = 0$

i.e., $x - 2y - 3z + 30 = 0$

Example 3. Test whether the plane x = 3 touches the sphere $x^2 + y^2 + z^2 = 9$.

Solution: The condition that the plane lx + my + nz = p to touch the sphere

$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ is

$$\frac{l(-u) + m(-v) + n(-w) - p}{\sqrt{l^2 + m^2 + n^2}} = \sqrt{u^2 + v^2 + w^2 - d}$$

i.e., $(lu + mv + nw + p)^2 = (l^2 + m^2 + n^2)(u^2 + v^2 + w^2 - d)$. (1)

Here $u = 0, v = 0, w = 0, l = 1, m = 0, n = 0, p = 3, d = -4$.

$(1) \Rightarrow (0+0+3)^2 = (1+0+0)(0+0+0+9)$

i.e., $3^2 = 9$, which is satisfied.

Hence the plane x = 3 touches the sphere $x^2 + y^2 + z^2 = 9$.

Example 4. Find the equation of the sphere which has its centre at (-1, 2, 3) and touches the plane 2x-y+2z = 6.

Solution: Let the equation of the sphere be

$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$.

Given: $-u = -1, \ -v = 2, \ -w = 3$ i.e. $u = 1, \ v = -2, \ w = -3$. (1)

$\therefore (1) \Rightarrow x^2 + y^2 + z^2 + 2x - 4y - 6z + d = 0$. (2)

To find d: Since the plane 2x-y+2z = 6 touches the sphere,whose centre is (-1, 2, 3). The radius of the sphere is equal to the length of the perpendicular drawn from the centre (-1, 2, 3) to the plane 2x-y+2z = 6.

Length of the perpendicular = $\dfrac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}$

$$= \frac{(2)(-1) + (-1)(2) + (2)(3) - 6}{\sqrt{4+1+4}} = \frac{-2 - 2 + 6 - 5}{\sqrt{9}}$$

$$= \frac{-4}{3} = r.$$

We know that $r = \sqrt{u^2 + v^2 + w^2 - d}$.

So $r^2 = u^2 + v^2 + w^2 - d$

i.e. $d = u^2 + v^2 + w^2 - r^2 = (-1)^2 + (2)^2 + (3)^2 - \left(\frac{-4}{3}\right)^2$

$$= 1 + 4 + 9 - \frac{16}{9} = 14 - \frac{16}{9} = \frac{110}{9}.$$

$(2) \Rightarrow x^2 + y^2 + z^2 + 2x - 4y - 6z + \frac{110}{9} = 0$

i.e. $9(x^2 + y^2 + z^2) + 18x - 36y - 54z + 110 = 0$.

This is the required equation of the sphere.

Example 5. Find the equation of the sphere with centre at (2, 3, 5), which touches the XOY plane.

Solution: Let centre $(2,3,5) = (x_1, y_1, z_1)$ and r is the radius of the required sphere.

Formula: Radius = perpendicular distance from (x_1, y_1, z_1) to the plane ax + by + cz +d = 0

$$= \pm \frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}.$$

So radius r = perpendicular distance from (2, 3, 5) to the plane z = 0

$$= \pm \frac{5}{\sqrt{0^2 + 0^2 + 1^2}} = \pm 5.$$

The required sphere is $(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 = r^2$

i.e. $(x-2)^2 + (y-3)^2 + (z-5)^2 = 5^2$

i.e. $x^2 - 4x + 4 + y^2 - 6y + 9 + z^2 - 10z + 25 = 25$

i.e. $x^2 + y^2 + z^2 - 4x - 6y - 10z + 13 = 0$.

Example 6. Find the equation of the sphere for which the circle $x^2+y^2+z^2+7y-2z+2=0$, $2x+3y+4z=8$ is a great circle.

Solution: The given circle is

$$S \equiv x^2+y^2+z^2+7y-2z+2=0,\ P \equiv 2x+3y+4z-8=0. \quad \text{—(1)}$$

Sphere through the above circle is of the form $S+\lambda P=0$

i.e. $x^2+y^2+z^2+7y-2z+2+\lambda(2x+3y+4z-8)=0$

i.e. $x^2+y^2+z^2+2\lambda x+(7+3\lambda)y+(-2+4\lambda)z+2-8\lambda=0$. —(2)

Centre of the sphere (2) is

$$(-u,-v,-w) = \left(-\left(\frac{2\lambda}{2}\right),-\left(\frac{7+3\lambda}{2}\right),-\left(\frac{-2+4\lambda}{2}\right)\right)=\left(-\lambda,\frac{-7-3\lambda}{2},1-2\lambda\right).$$

If the circle (1) is a great circle of the sphere (2), then the centre of (2) lies on the plane P = 0

$$\Rightarrow\ 2(-\lambda)+3\left(\frac{-7-3\lambda}{2}\right)+4(1-2\lambda)-8=0$$

$$\Rightarrow\ -2\lambda-\frac{21}{2}-\frac{9\lambda}{2}+4-8\lambda-8=0$$

$$\Rightarrow\ -\frac{29\lambda}{2}=\frac{29}{2}\Rightarrow \lambda=-1.$$

Putting the value of λ in equation (2), the requiered equation of the sphere is

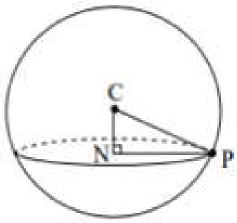
$$x^2+y^2+z^2-2x+4y-6z+10=0.$$

Example 7. Find the centre and radius of the circle in which the sphere $x^2+y^2+z^2+2y+4z-11=0$ is cut by the plane $x+2y+2z+15=0$.

Solution: The centre and radius of the sphere $x^2+y^2+z^2+2y+4z-11=0$ —(1) are C(0, -1, -2) and $CP=\sqrt{0^2+1^2+2^2+11}=4$.

Let N be the foot of the perpendicular from C on the plane $x+2y+2z+15=0$. —(2)

Then $CN=\dfrac{0+2(-1)+2(-2)+15}{\sqrt{1^2+2^2+2^2}}=3$.



$\therefore$ The radius of the circle $=\sqrt{CP^2-CN^2}=\sqrt{4^2-3^2}$
$=\sqrt{7}$.

The centre of the circle is N.

Now CN is the perpendicular to the plane (2). So its d.r.'s are 1, 2, 2.

Also CN passes through C(0, -1, -2). So the equations of the line CN are

$$\frac{x-0}{1}=\frac{y+1}{2}=\frac{z+2}{2}=r\ \text{(say)}.$$

So any point on this line is (r, 2r - 1, 2r - 2). If this point is N, then it lies on the plane (2).

$\therefore\quad r+2(2r-1)+2(2r-2)+15=0$

i.e., $9r+9=0$

i.e., $r=-1$.

$\therefore$ The coordinates of N are (-1, 2(-1)-1, 2(-1)-2)

i.e., **(-1, -3, -4)**

Thus the centre and radius of the circle are (-1, -3, -4) and $\sqrt{7}$.

Example 8. A sphere touches the plane $x - 2y - 2z - 7 = 0$ in the point (3, –1, –1) and passes through the point (1, 1, –3). Find its equation.

Solution: The equation of the point sphere with centre at the point (3, –1, –1) is

$$(x - 3)^2 + (y + 1)^2 + (z + 1)^2 = 0$$

i.e., $$x^2 + y^2 + z^2 - 6x + 2y + 2z + 11 = 0 \qquad (1)$$

The required sphere touches the plane

$$x - 2y - 2z - 7 = 0 \qquad (2)$$

at (3, -1, -1)

Therefore It contains the point circle of intersection of sphere (1) and plane (2).

Hence the equation of the required sphere is of the form.

$$x^2 + y^2 + z^2 - 6x + 2y + 2z + 11 + \lambda(x - 2y - 2z - 7) = 0$$

and passes through (1, 1, -3)

$$\therefore \quad 1^2 + 1^2 + (-3)^2 - 6.1 + 2.1 + 2(-3) + 11 + \lambda(1 - 2.1 - 2(-3) - 7) = 0.$$

$$12 + \lambda(-2) = 0$$

$$\lambda = 6$$

$\therefore$ The required sphere is

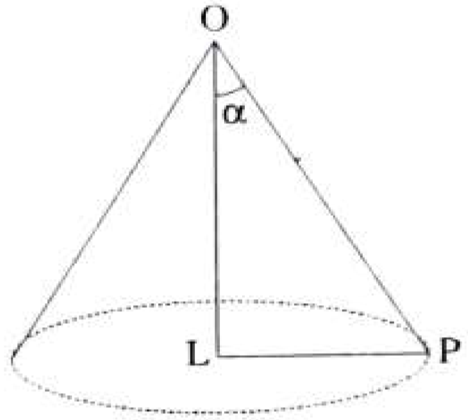
$$x^2 + y^2 + z^2 - 6x + 2y + 2z + 11 + 6(x - 2y - 2z - 7) = 0$$

i.e., $$x^2 + y^2 + z^2 - 10y - 10z - 31 = 0$$

LECTURE NO.:- 17

RIGHT CIRCULAR CONE

A right circular cone is a surface generated by a line which passes through a fixed point and makes a constant angle with a fixed line through the fixed point.



The fixed point is the *vertex* of the cone: the fixed line is called the *axis* and the constant angle the *semi-vertical angle* of the cone.

Every section of a right circular cone by a plane perpendicular to its axis is a circle. To prove it, suppose that a plane perpendicular to the axis OL of the right circular cone whose semi-vertical angle is α, meets it at L. If P be any point on the section, OLP
If P be any point on the section, OLP
$= 90°$, since OL is perpendicular to the plane.

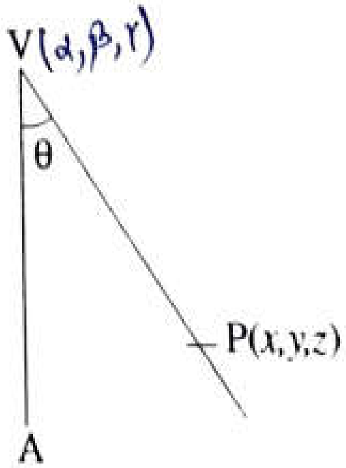
In triangle OLP, $PL = OL \tan \alpha$ and since OL is a fixed quantity, being the distance of the plane from the vertex, PL is constant for every position of P on the section.

Therefore, the section is a circle with centre at L and radius PL.

EQUATION OF A RIGHT CIRCULAR CONE

To find the equation of the right circular cone with its vertex at the point (α, β, γ), semi-vertical angle θ and whose axis is the line

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}. \qquad \text{..... (1)}$$



Let $V(\alpha, \beta, \gamma)$ be the vertex and VA the axis of the cone. Let $P(x, y, z)$ be any point on the surface of the cone, so that VP is a generator of the cone. The angle between VP and the axis VA is θ; the direction ratios of VP are $x-\alpha$, $y-\beta$, $z-\gamma$ and consequently

$$\cos\theta = \frac{l(x-\alpha) + m(y-\beta) + n(z-\gamma)}{\sqrt{(l^2+m^2+n^2)}\ \sqrt{\{(x-\alpha)^2}}$$

This gives

$$[l(x-\alpha)+m(y-\beta)+n(z-\gamma)]^2$$
$$=\cos^2\theta(l^2+m^2+n^2)\,[(x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2], \qquad \ldots\ldots (2)$$

which is the required equation of the right circular cone.

Cor. 1. If the vertex is the origin, the equation of the cone becomes

$$(lx+my+nz)^2=(l^2+m^2+n^2)(x^2+y^2+z^2)\cos^2\theta \qquad \ldots\ldots (3)$$

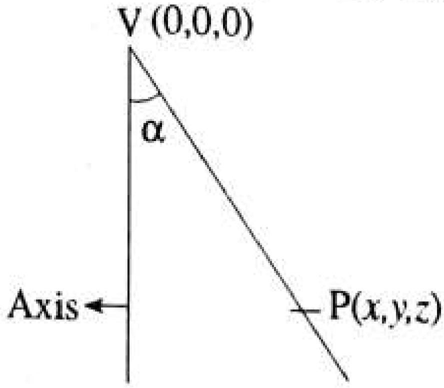
Cor. 2. If the vertex is the origin and the z-axis is the axis of the cone, its equation is ($\because l=0, m=0, n=1$)

$$z^2=(x^2+y^2+z^2)\cos^2\theta,$$
$$\Rightarrow x^2+y^2=z^2\tan^2\theta. \qquad \ldots\ldots (4)$$

Example 1. Find the equation to the right circular cone with vertex at the origin, axis the line $\frac{x}{2}=\frac{y}{-4}=\frac{z}{3}$ and which passes through the point (1, 1, 2).

Solution. Let α be the semi vertical angle of the cone. **[MNIT, 2003]**

The d.r's of VP are 1, 1, 2. The d.r's of the axis are 2, –4, 3.



$$\therefore \quad \cos\alpha=\frac{2-4+6}{\sqrt{6}\sqrt{29}}=\frac{4}{\sqrt{6}\sqrt{29}}$$

If (x, y, z) be any point on the surface of the right circular cone,

$$\cos\alpha=\frac{2x-4y+3z}{\sqrt{29}\sqrt{x^2+y^2+z^2}}$$

$$\Rightarrow \quad \frac{4}{\sqrt{6}\sqrt{29}}=\frac{2x-4y+3z}{\sqrt{29}\sqrt{x^2+y^2+z^2}}$$

$$\Rightarrow \quad 16(x^2+y^2+z^2)=6(2x-4y+3z)^2$$

$$\Rightarrow \quad 8(x^2+y^2+z^2)=3(4x^2+16y^2+9z^2-16xy-24yz+12zx)$$

$$\Rightarrow \quad 4x^2+40y^2+19z^2-48xy-72yz+36zx=0.$$

This is the equation of the right circular cone.

Example 2. Find the equation of the right circular cone generating by three straight lines drawn from the origin to cut the circle through three points (1, 2, 2), (2, 1, – 2) and (2, – 2, 1).

Solution. Here the vertex of the cone is origin (0, 0, 0).

If α is the semi-vertical angle of the cone and l, m, n be the d.r.'s of the axis of the cone, we have

$$\cos\alpha = \frac{l + 2m + 2n}{3\sqrt{l^2 + m^2 + n^2}} \quad \text{...(1)}$$

$$\cos\alpha = \frac{2l + m - 2n}{3\sqrt{l^2 + m^2 + n^2}} \quad \text{....(2)}$$

$$\cos\alpha = \frac{2l - 2m + n}{3\sqrt{l^2 + m^2 + n^2}} \quad \text{....(3)}$$

(1) and (2) $\Rightarrow$ $l + 2m + 2n = 2l + m - 2n$

$\Rightarrow -l + m + 4n = 0$ ....(5)

(2) and (3) $\Rightarrow$ $2l + m - 2n = 2l - 2m + n$

$\Rightarrow 3m - 3n = 0$

$\therefore\ m = n$ ....(6)

from (5), $-l + 5n = 0 \Rightarrow l = 5n$

$$\therefore \quad \frac{l}{5} = \frac{m}{1} = \frac{n}{1} = \frac{1}{\sqrt{27}}$$

From (1), $\cos\alpha = \dfrac{l + 2m + 2n}{3\sqrt{l^2 + m^2 + n^2}}$

$$\Rightarrow \quad \cos\alpha = \frac{5 + 2 + 2}{3\sqrt{27}} = \frac{3}{\sqrt{27}} = \frac{1}{\sqrt{3}}.$$

If (x, y, z) be any point on the right circular cone,

$$\cos\alpha = \frac{lx + my + nz}{\sqrt{l^2 + m^2 + n^2}\sqrt{x^2 + y^2 + z^2}}$$

$$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{5x + y + z}{\sqrt{27}\sqrt{x^2 + y^2 + z^2}}$$

$$9(x^2 + y^2 + z^2) = (5x + y + z)^2$$

$$\Rightarrow \quad 16x^2 - 8y^2 - 8z^2 + 10xy + 2yz + 10zx = 0.$$

$$\Rightarrow \quad 8x^2 - 4y^2 - 4z^2 + 5xy + yz + 5zx = 0.$$

Questions for Practice:

1. Find the equation of the right circular cone with vertex (2, 3, 1), axis parallel to the line $\frac{x}{-1} = \frac{y}{2} = \frac{z}{1}$ and one of its generators having direction ratios 1, –1, 1.
2. Find the equations of the right circular cones which contain the three coordinate axes as generators.
3. If a right circular cone has three mutually perpendicular generators, show that the semi-vertical angle is $\tan^{-1}(\sqrt{2})$.

LECTURE NO.:- 18

RIGHT CIRCULAR CYLINDER

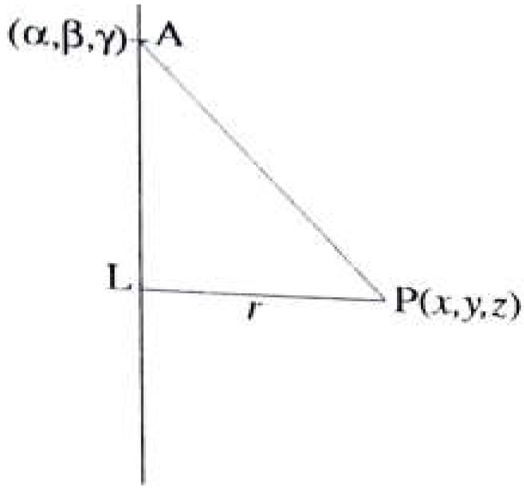
A right circular cylinder is a surface generated by a line which intersects a fixed circle and is perpendicular to the plane of the circle. The circle is called the guiding circle and the normal to the plane of the guiding circle through its centre is called the axis of the cylinder.

Section of a right circular cylinder by any plane perpendicular to its axis is called a normal section. All normal sections of a right circular cylinder are circles, having the same radius. This radius is called the radius of the cylinder. The perpendicular distance of any point on the right circular cylinder, from the axis of the cylinder, is equal to the radius of the cylinder. This fact helps-in obtaining the equation to a right circular cylinder.

EQUATION OF A RIGHT CIRCULAR CYLINDER

To find the equation of the right circular cylinder of radius r and whose axis is the line

$$\frac{x-\alpha}{l}=\frac{y-\beta}{m}=\frac{z-\gamma}{n}. \qquad \text{..... (1)}$$



Let AL be the axis and let P(x, y, z) be the any point on the surface of right circular cylinder. Draw PL, perpendicular from P to the axis, meeting it at L.

Now

AL = Projection of the line joining the points A (α, β, γ) and P (x, y, z) upon the line AL, with direction ratios l, m, n.

$$\therefore \quad AL = \frac{(x-\alpha)\,l + (y-\beta)\,m + (z-\gamma)\,n}{\sqrt{(l^2+m^3+n^2)}}$$

Since triangle ALP is right angled at L. $AP^2 = AL^2 + r^2$

$$\Rightarrow (x-\alpha)^2+(y-\beta)^2+(z-\gamma)^2 = \frac{[(x-\alpha)\,l+(y-\beta)\,m+(z-\gamma)\,n]^2}{(l^2+m^2+n^2)} + r^2.$$

This is the required equation of the right circular cylinder.

Example 1. Find the equation of the right circular cylinder whose axis is $x = 2y = -z$ and radius 4.

Solution. The axis is given to be $\frac{x}{2} = \frac{y}{1} = \frac{z}{-2}$.

Let $P(x, y, z)$ be any point on the surface of the right circular cylinder. The perpendicular distance of P from the axis is equal to 4.

$$\therefore \quad 16 = (x^2 + y^2 + z^2) - \left[\frac{2(x-0) + 1(y-0) - 2(z-0)}{3}\right]^2$$

$$144 = 9(x^2 + y^2 + z^2) - (2x + y - 2z)$$

$$= 9(x^2 + y^2 + z^2) - (4x^2 + y^2 + 4z^2 + 4xy - 4yz - 8zx]$$

or $\quad 5x^2 + 8y^2 + 5z^2 - 4xy + 4yz + 8zx = 144.$

This is the equation of the right circular cylinder.

Example 2. Obtain the equation of the right circular cylinder described on the circle through the three points (1, 0, 0), (0, 1, 0) and (0, 0, 1) as guiding circle.

Solution. Taking fourth point as (0, 0, 0), the equation of the sphere passing through the four points (0, 0, 0), (1, 0, 0), (0, 1, 0) and (0, 0, 1) is to be determined.

If equation of the sphere is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$, then passing through the points (0, 0, 0), (1, 0, 0), (0, 1, 0) and (0, 0, 1), the values $d = 0$, $u = -\frac{1}{2}$, $v = -\frac{1}{2}$ and $w = -\frac{1}{2}$ are obtained. Hence the equation of the sphere becomes

$$x^2 + y^2 + z^2 - x - y - z = 0. \qquad \text{..... (1)}$$

Equation of the plane passing through the given three points (1, 0, 0), (0, 1, 0) and (0, 0, 1) is $\quad x + y + z = 0.$ ..... (2)

Equations (1) and (2) together represent a circle. Now we shall find the centre and radius of this circle.

Centre of the sphere is $C\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ and its radius $= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$.

Let P be the centre of the circle. P is the foot of the $\perp$ from C to (2).

Equations of CP are

$$\frac{x - 1/2}{1} = \frac{y - 1/2}{1} = \frac{z - 1/2}{1} \ (= k), \text{ say} \qquad \text{....(3)}$$

Any point on this line is $\left(k + \frac{1}{2}, k + \frac{1}{2}, k + \frac{1}{2}\right)$.

Let P be $\left(k + \frac{1}{2}, k + \frac{1}{2}, k + \frac{1}{2}\right)$ for some k. It lies on the plane (2).

$$\therefore \quad \left(k + \frac{1}{2}\right) + \left(k + \frac{1}{2}\right) + \left(k + \frac{1}{2}\right) = 1$$

$$\Rightarrow \quad 3k - \frac{1}{2} \Rightarrow k = -\frac{1}{6}.$$

$$\therefore \quad \text{P is } \left(\frac{-1}{6} + \frac{1}{2}, \frac{-1}{6} + \frac{1}{2}, \frac{-1}{6} + \frac{1}{2}\right) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

Radius of the circle is given by

$$(\text{radius})^2 = \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{\frac{1}{2} + \frac{1}{2} + \frac{1}{2} - 1}{\sqrt{3}}\right)^2$$

$$= \frac{3}{4} - \frac{1}{12} = \frac{8}{12} = \frac{2}{3}$$

$\Rightarrow$ radius of the circle $= \sqrt{\frac{2}{3}}$.

We now find the equaiton of the right circular cylinder. Axis has d.r's 1, 1, 1. If Q(x, y, z) be any point on the cylinder,

$$\left(\frac{\sqrt{2}}{\sqrt{3}}\right)^2 = \left[\left(x-\frac{1}{2}\right)^2 + \left(y-\frac{1}{2}\right)^2 + \left(z-\frac{1}{2}\right)^2\right] - \left[\frac{\left(x-\frac{1}{2}\right)+\left(y-\frac{1}{2}\right)+\left(z-\frac{1}{2}\right)}{\sqrt{3}}\right]^2$$

$$\frac{2}{3} = \left[\left(x-\frac{1}{2}\right)^2 + \left(y-\frac{1}{2}\right)^2 + \left(z-\frac{1}{2}\right)^2\right] - \frac{1}{3}\left[x+y+z-\frac{3}{2}\right]^2$$

$$\frac{2}{3} = \left(x^2+y^2+z^2-x-y-z+\frac{3}{4}\right) - \frac{1}{3}\left[x^2+y^2+2xy+z^2-3z+\frac{9}{4}+2xz-3x+2yz-3y\right]$$

$$2 = 3x^2+3y^2+3z^2-3x-3y-3z+\frac{9}{4}-x^2-y^2-z^2+3x+3y+3z-\frac{9}{4} - 2xy-2yz-2zx.$$

$\Rightarrow$ $2x^2+2y^2+2z^2-2xy-2yz-2zx = 2$

or, $x^2+y^2+z^2-xy-yz-zx = 1.$

This is the equation of the right circular cylinder.

Questions for practice:

1. Find the equation of the right circular cylinder whose guiding curve is the circle
   $x^2+y^2+z^2=9, \quad x-y+z=3.$
2. The axis of a right circular cylinder of radius 2 is
   $\frac{x-1}{2} = \frac{y}{3} = \frac{z-3}{1}$.
   Find its equation.
3. Find the equation of the right circular cylinder of radius 2 whose axis is the line
   $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{2}$.
4. The radius of a normal section of a right circular cylinder is 2 units; the axis lies along the line
   $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z-2}{5}$.
   Find its equation.